

Computer Fundamentals and Programming

CIS-101L



FALL-2025

CLO Marks	
Obtained Marks	
Lab Engineer Signature	
Comments	

Project Report

Group members

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Semester duration 8 Sep to 26 Dec, 2025	
Section: B	Group : B-4
Project Title: Hotel Room Booking System	Project report Submission date: December-16-2025
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Department of Electrical Engineering

Computer Fundamentals and Programming

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Project Title: Hotel Room Booking System

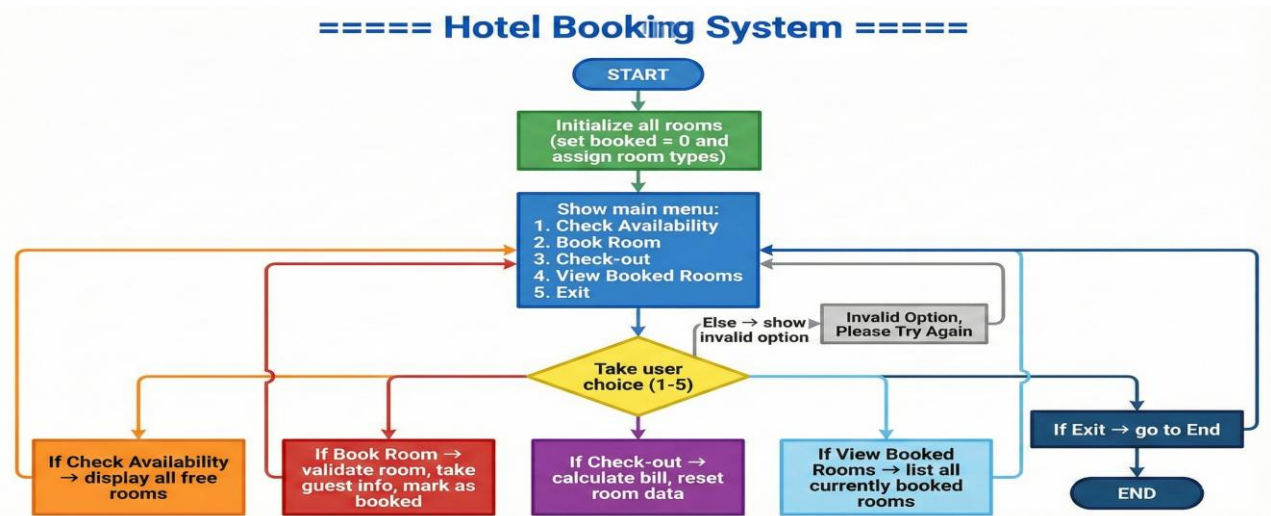
Description:

- Room Types: **Single, Double, Suite** with prices.
- Check room availability.
- Book Room: Guest Name, CNIC, Phone, Days of Stay.
- Check-out: Input Room No., Calculate Rent (Price Days).
- Add **10%** service tax.
- View all booked rooms.

Abstract:

The Hotel Room Booking System is a simple program to manage hotel rooms easily. It has three types of rooms: Single, Double, and Suite, each with its own price. The system shows which rooms are free, so it is easy to know what can be booked. Guests can book a room by giving their name, CNIC, phone number, and number of days they will stay. All the information is saved in arrays, which keeps it organized. When a guest checks out, the program calculates the total cost by multiplying the room price with days stayed and adds 10% service tax. It also makes sure a room cannot be booked twice and that the room number is valid. Users can see all booked rooms with guest details anytime. The menu is simple and easy to use. Each part of the program has its own function to keep the code clean. This system saves time for staff and reduces mistakes. It makes booking and checkout very smooth. Overall, it is a simple, useful, and easy-to-use hotel management system.

FLOW CHART :



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Parts of Code With Descriptions :

(1) Setup & Room Data

Code:

```
#include <iostream>
#include <string>
using namespace std;

const int TOTAL_ROOMS = 45;

int roomType[TOTAL_ROOMS];
bool booked[TOTAL_ROOMS];
string guestName[TOTAL_ROOMS];
string guestCNIC[TOTAL_ROOMS];
string guestPhone[TOTAL_ROOMS];
int daysStay[TOTAL_ROOMS];
```

Description:

This section sets up the basic structure of the program. We include libraries for input/output and text handling. The constant TOTAL_ROOMS defines the total number of rooms in the hotel as 45. Arrays are created to store information about each room, including its type, whether it is booked, guest name, CNIC, phone, and days of stay. Using arrays makes it easy to manage all rooms together.

(2) Room Price Function

Code:

```
int getPrice(int type) {
    if (type == 1) return 3000;
    else if (type == 2) return 5000;
    else if (type == 3) return 8000;
    return 0;
}
```

Description:

This function returns the price of a room based on its type. Single rooms cost 3000, Double rooms 5000, and Suites 8000. Using this function keeps the code organized because we don't need to write the prices again and again.

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(3) Initialize Rooms Function

Code:

```
void initializeRooms() {
    for (int i = 0; i < TOTAL_ROOMS; i++) {
        booked[i] = 0;
        if (i < 15)
            roomType[i] = 1;
        else if (i < 30)
            roomType[i] = 2;
        else
            roomType[i] = 3;
    }
}
```

Description:

This function sets all rooms to free when the program starts. It also assigns types for each room: first 15 are Single, next 15 are Double, and last 15 are Suites. This ensures the program starts with clean, correct data.

(4) Check Availability Function

Code:

```
void checkAvailability() {
    cout << endl << "--- Available Rooms ---" << endl;
    bool anyFree = 0;
    cout << "Room No\t\tType\t\tPrice" << endl;
    cout << "-----" << endl;
    for (int i = 0; i < TOTAL_ROOMS; i++) {

        if (booked[i] == 0) {
            anyFree = 1;
            string typeName;
            if (roomType[i] == 1) typeName = "Single";
            else if (roomType[i] == 2) typeName = "Double";
            else typeName = "Suite";
            cout << "Room " << i + 1 << "\t\t" << typeName << "\t\tRs. " << getPrice(roomType[i]) << endl;
        }
    }
    if (anyFree == 0)
        cout << "No rooms available." << endl;
}
```

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Description:

This function shows all the rooms that are currently free. It checks each room and if the room is not booked, it displays its number, type, and price. If no rooms are free, it informs the user. This helps the staff or guest know which rooms they can book.

(5) Book a Room Function

Code:

```
void bookRoom() {  
  
    cout << endl << "Room Types:" << endl;  
    cout << " 1 - 15 : Single (Rs. 3000)" << endl;  
    cout << "16 - 30 : Double (Rs. 5000)" << endl;  
    cout << "31 - 45 : Suite (Rs. 8000)" << endl;  
  
    int room;  
    cout << endl << "Enter room number to book (1-45): ";  
    cin >> room;  
  
    if (room < 1 || room > TOTAL_ROOMS) {  
        cout << "Invalid room number." << endl;  
    }  
    else {  
        room--;  
        if (booked[room] == 1) {  
            cout << "Room already booked." << endl;  
        }  
        else {  
            cin.ignore();  
            cout << "Enter Guest Name: ";  
            getline(cin, guestName[room]);  
  
            cout << "Enter CNIC: ";  
            cin >> guestCNIC[room];  
  
            cout << "Enter Phone: ";  
            cin >> guestPhone[room];  
  
            cout << "Enter number of days: ";  
            cin >> daysStay[room];  
  
            booked[room] = 1;  
            cout << "Room booked successfully!" << endl;  
        }  
    }  
}
```

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Description:

This function allows the user to book a room. It first shows the types of rooms and prices. Then it asks for the room number and checks if it is valid and free. If the room is available, it collects guest details including name, CNIC, phone number, and days of stay. Finally, it marks the room as booked. This ensures that bookings are done in an organized way without errors.

(6) Check-Out Function

Code:

```
void checkOut() {

    int room;
    cout << endl << "Enter room number for check-out: ";
    cin >> room;

    if (room < 1 || room > TOTAL_ROOMS) {
        cout << "Invalid room number." << endl;
    }

    else {
        room--;
        if (booked[room] == 0) {
            cout << "Room is not booked." << endl;
        }
        else {
            int pricePerDay = getPrice(roomType[room]);
            int baseAmount = pricePerDay * daysStay[room];
            float tax = baseAmount * 0.10;
            float totalBill = baseAmount + tax;

            cout << endl << "--- BILL ---" << endl;
            cout << "Guest Name: " << guestName[room] << endl;
            cout << "Price per Day: Rs. " << pricePerDay << endl;
            cout << "Days Stayed: " << daysStay[room] << endl;
            cout << "Base Rent: Rs. " << baseAmount << endl;
            cout << "Service Tax (10%): Rs. " << tax << endl;
            cout << "Total Amount: Rs. " << totalBill << endl;
            booked[room] = 0;

            cout << "Check-out completed." << endl;
        }
    }
}
```

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Description:

This function allows a guest to check out of a room. It asks for the room number, checks if it is valid and booked, then calculates the total bill including a 10% service tax. It displays the guest's name, price, days stayed, and total amount to pay. Finally, it frees the room, making the room available for the next guest.

(7) View Booked Rooms Function

Code:

```
void viewBooked() {
    cout <<endl<< " --- Booked Rooms ---"<<endl;
    bool any = false;

    for (int i = 0; i < TOTAL_ROOMS; i++) {
        if (booked[i] == 1) {
            any = true;

            string typeName;
            if (roomType[i] == 1)
                typeName = "Single";
            else if (roomType[i] == 2)
                typeName = "Double";
            else
                typeName = "Suite";

            cout <<endl<< "Room " << (i + 1) << " (" << typeName << ")"<<endl;
            cout << "Guest: " << guestName[i] << endl;
            cout << "CNIC: " << guestCNIC[i] << endl;
            cout << "Phone: " << guestPhone[i] << endl;
            cout << "Days: " << daysStay[i] << endl;
        }
    }

    if (any==0)
        cout << "No rooms are currently booked."<<endl;
}
```

Description:

This function shows all rooms that are currently booked. For each booked room, it displays the room number, type, guest name, CNIC, phone number, and days of stay. If no rooms are booked, it informs the user. This helps the hotel staff keep track of all occupied rooms and guest details.

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(8) Main Function (Menu)

Code:

```
int main() {
    initializeRooms();

    int choice;

    do {
        cout <<endl << "==== Hotel Booking System =====<<endl;
        cout << "1. Check Availability"<<endl;
        cout << "2. Book Room"<<endl;
        cout << "3. Check-out"<<endl;
        cout << "4. View Booked Rooms"<<endl;
        cout << "5. Exit"<<endl;
        cout << "Choose an option: ";
        cin >> choice;

        switch (choice) {
            case 1: checkAvailability();
            break;
            case 2: bookRoom();
            break;
            case 3: checkOut();
            break;
            case 4: viewBooked();
            break;
            case 5: cout << "Exiting program."<<endl;
            break;
            default: cout << "Invalid option."<<endl;
        }
    } while (choice != 5);

    return 0;
}
```

Description:

The main function runs the hotel booking system. It first initializes all rooms. Then it shows a menu repeatedly, allowing the user to choose actions like checking room availability, booking a room, checking out, viewing booked rooms, or exiting. Based on the user's choice, it calls the corresponding function. The menu continues until the user chooses to exit. This provides a simple, interactive interface for managing the hotel system.

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OUTPUTS :

1.Check Availability :

```
===== Hotel Booking System =====
1. Check Availability
2. Book Room
3. Check-out
4. View Booked Rooms
5. Exit
Choose an option: 1

--- Available Rooms ---
Room No      Type      Price
-----
Room 1       Single     Rs. 3000
Room 2       Single     Rs. 3000
Room 3       Single     Rs. 3000
Room 4       Single     Rs. 3000
Room 5       Single     Rs. 3000
Room 6       Single     Rs. 3000
Room 7       Single     Rs. 3000
Room 8       Single     Rs. 3000
Room 9       Single     Rs. 3000
Room 10      Single     Rs. 3000
Room 11      Single     Rs. 3000
Room 12      Single     Rs. 3000
Room 13      Single     Rs. 3000
Room 14      Single     Rs. 3000
Room 15      Single     Rs. 3000
Room 16      Double     Rs. 5000
Room 17      Double     Rs. 5000
Room 18      Double     Rs. 5000
Room 19      Double     Rs. 5000
Room 20      Double     Rs. 5000
Room 21      Double     Rs. 5000
Room 22      Double     Rs. 5000
Room 23      Double     Rs. 5000
Room 24      Double     Rs. 5000
Room 25      Double     Rs. 5000
Room 26      Double     Rs. 5000
Room 27      Double     Rs. 5000
Room 28      Double     Rs. 5000
Room 29      Double     Rs. 5000
Room 30      Double     Rs. 5000
Room 31      Suite      Rs. 8000
Room 32      Suite      Rs. 8000
Room 33      Suite      Rs. 8000
Room 34      Suite      Rs. 8000
Room 35      Suite      Rs. 8000
Room 36      Suite      Rs. 8000
Room 37      Suite      Rs. 8000
Room 38      Suite      Rs. 8000
Room 39      Suite      Rs. 8000
Room 40      Suite      Rs. 8000
Room 41      Suite      Rs. 8000
Room 42      Suite      Rs. 8000
Room 43      Suite      Rs. 8000
Room 44      Suite      Rs. 8000
Room 45      Suite      Rs. 8000
```

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2.Book Room :

```
===== Hotel Booking System =====
1. Check Availability
2. Book Room
3. Check-out
4. View Booked Rooms
5. Exit
Choose an option: 2

Room Types:
1 - 15 : Single (Rs. 3000)
16 - 30 : Double (Rs. 5000)
31 - 45 : Suite (Rs. 8000)

Enter room number to book (1 - 45): 2
Enter Guest Name: Muhammad Sufyan
Enter CNIC: 33104-5555555-5
Enter Phone: 0300-000000000
Enter number of days: 3
Room booked successfully!
```

3.Check Out :

```
===== Hotel Booking System =====
1. Check Availability
2. Book Room
3. Check-out
4. View Booked Rooms
5. Exit
Choose an option: 3

Enter room number for check-out: 2

--- BILL ---
Guest Name: Muhammad Sufyan
Price per Day: Rs. 3000
Days Stayed: 3
Base Rent: Rs. 9000
Service Tax (10%): Rs. 900
Total Amount: Rs. 9900
Check-out completed.
```

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4.View Booked Rooms :

```
===== Hotel Booking System =====
1. Check Availability
2. Book Room
3. Check-out
4. View Booked Rooms
5. Exit
Choose an option: 4

--- Booked Rooms ---

Room 2 (Single)
Guest: Muhammad Sufyan
CNIC: 33104-5555555-5
Phone: 0300-0000000
Days: 3
```

5.Exit :

```
===== Hotel Booking System =====
1. Check Availability
2. Book Room
3. Check-out
4. View Booked Rooms
5. Exit
Choose an option: 5
Exiting program.
```

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Conclusion:

This project is a **Hotel Room Booking System** made in C++. It manages 45 rooms in total, which include Single, Double, and Suite rooms. The program uses arrays to store room type, booking status, guest name, CNIC, phone number, and number of days stayed. When the program starts, all rooms are set as **free**. The system can show all available rooms along with their type and price. A user can **book** any room that is not already taken. During booking, the program asks for guest information. It also prevents **double booking** and stops the user from entering a wrong room number. The system allows the user to **check out** a guest as well. While checking out, it calculates the total bill by multiplying price per day with days stayed and then adding a 10% service tax. After the check-out, the room becomes free again for new guests. The user can also view all the rooms that are currently **booked**. The main menu keeps appearing until the user exits. Overall, this code helps manage hotel rooms in a very simple, clear, and useful way.